

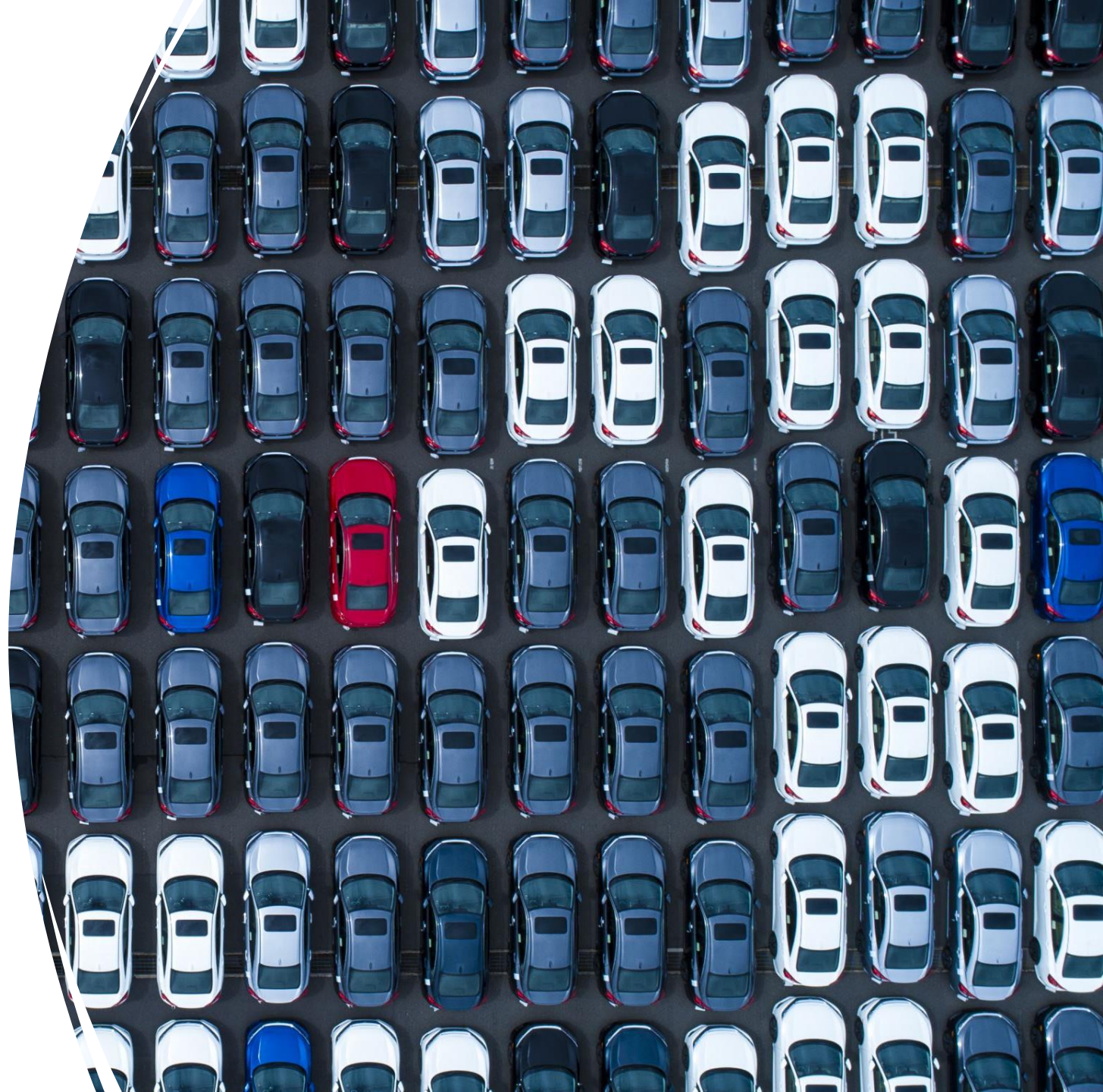
Optimizing Rental Car Revenue: A Hands-On Introduction to Revenue Management

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Chris Garcia

University of Mary Washington

Polyad Decision Sciences, LLC



Agenda

Overview of Revenue Management

Modeling Scenario 1: RM in a Single-Location Rental Car Business

Modeling Scenario 2: Basic Multi-Location Rental Car RM

Modeling Scenario 3: Multiple Locations with Realistic Transfers

Modeling Scenario 4: Incorporating Dynamic Pricing and Fluctuating Demand

Ethical Use of Revenue Management

Questions and Discussion



University of
Mary Washington

Introduction

- My work at UMW
 - <https://cgarcia.info/>
- My work at PDS
 - <https://polyad.ai/>

Overview of Revenue Management

- If you have ever bought plane tickets, you have probably noticed the impact timing has on price:
 - One person paid \$200 for their seat reserved 6 months in advance
 - The person next to them paid \$3,300 for their seat reserved the day before
- Why such a large difference for seemingly the same good?
- We see similar phenomena occurring in many other industries also:
 - Booking a hotel
 - Reserving a rental car
 - Scheduling tee time at a golf course
 - List goes on
- What is behind this, and how does it work?
- Answer: Revenue Management

Overview of Revenue Management (cont.)



Revenue Management (RM)

- The science of maximizing revenues attained with existing capacity through the use of advanced analytics

Goal of Revenue Management

- Sell the right product
- To the right customer
- At the right time
- Through the right channel
- At the right price

Each of these “right” unknowns must be determined

Where Can Revenue Management be Used?

- A wide variety of industries, ranging from car rentals to airlines to air conditioner repair
- Generally, works best when the following conditions hold:
 - Fixed capacity
 - A rental car business has 20 luxury cars and 40 compact cars
 - A plane has 50 seats
 - Perishable goods or services
 - When a day passes, any unrented cars = lost revenue
 - When a plane takes off, unsold seats = lost revenue
 - Reservation of resources/services ahead of time
 - This can include last-minute/spontaneous sales

Key Revenue Management Activities

1

Segment the Market

Determine the different kinds of customers and what they want from your product/service

2

Determine the Right Prices

Estimate the response from each segment to different prices and determine best prices for each segment

3

Forecast Demand

Estimate the demand from each segment over a defined time horizon

4

Optimize Capacity Allocation

Determine the right number of units to allocate to each segment and time period over the defined horizon

Single-Location Rental Car Revenue Management

Scenario 1

Car Class	Rent Day	Return Day	Demand	Price
Comp.	1	2	16	50
Comp.	1	3	20	102
Comp.	1	4	17	129
Midsize	1	2	12	63
Midsize	1	3	21	110
Midsize	1	4	21	180

Scenario 1 - Introduction

- In this scenario we have a single-location rental car business
 - Three car classes: Compact (35), Midsize (85), and Luxury (50)
 - Rental durations ranging from 1-3 days
 - Facilitate substitutions (upgrades)
 - 7 days of forecasted data
 - Realistically, 45-60 is common
- Each row is a booking slot
- How do we allocate our fleet to the booking slots to maximize revenue?

Scenario 1 – Framing

Known Entities

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)

S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon

p_k = Price per instance of car rented for slot k

d_k = Demand for slot k

D_j = Set of booking slots with cars departing (renting out) on day j

A_j = Set of booking slots with cars arriving (returning) on day j

I_{h0} = Initial inventory of car class h

$v_{hk} = 1$ if car class h can satisfy demand for slot k , 0 otherwise

Decision Variables

x_{hk} = Number of cars in class h used to fill slot k

I_{hj} = Inventory of car class h at end of business day j

Scenario 1 - Modeling

- Objective (1): Maximize total revenue
 - (Slot rental price × Cars assigned) summed over all car classes and slots
 - $\max \sum_{h \in C} \sum_{k \in S} p_k x_{hk}$
- Constraint (2): For each slot, ensure we don't allocate more cars than demand
 - Total cars allocated to a slot = sum of cars from all classes allocated to it
 - $\sum_{h \in C} x_{hk} \leq d_k \quad \forall k \in S$

Scenario 1 – Modeling (cont.)

- Constraint (3): For each slot, ensure only allowable car classes are allocated
 - To do this, ensure allocation of invalid class h to slot k is 0
 - Use the "Big M" - a large constant
 - $x_{hk} \leq Mv_{hk} \quad \forall h \in C; k \in S$
 - If $v_{hk} = 0$ then x_{hk} is forced to be 0, otherwise it can be large (constrained by other constraints, of course)
 - For solving efficiency, we want as small a value of M as possible
 - We can obtain a tight M by letting $M = \max(\text{Demand})$

Scenario 1 – Car Inventory Flow Modeling

- Numerous incompatible slots where a car cannot be assigned to both
 - To account for this, we model the flow of car inventory
- Constraint (4): For a given car class h on day j :
 - Ending Inventory j = Previous Day's Ending Inventory – Cars Rented Out Day j + Cars Returned Day j
 - $$I_{hj} = I_{hj-1} - \sum_{k \in D_j} x_{hk} + \sum_{k \in A_j} x_{hk} \quad \forall h \in C; j = 1.. \Delta$$
- Constraint (5): Ensure positive integer domains for all variables

$$I_{hj}, x_{hk} \geq 0 \text{ and integer } \forall h \in C; i \in L; j = 1.. \Delta; k \in S$$

Try It Out!

Let's go through
section 1 together in
the code notebook

Basic Multi-Location Rental Car Revenue Management

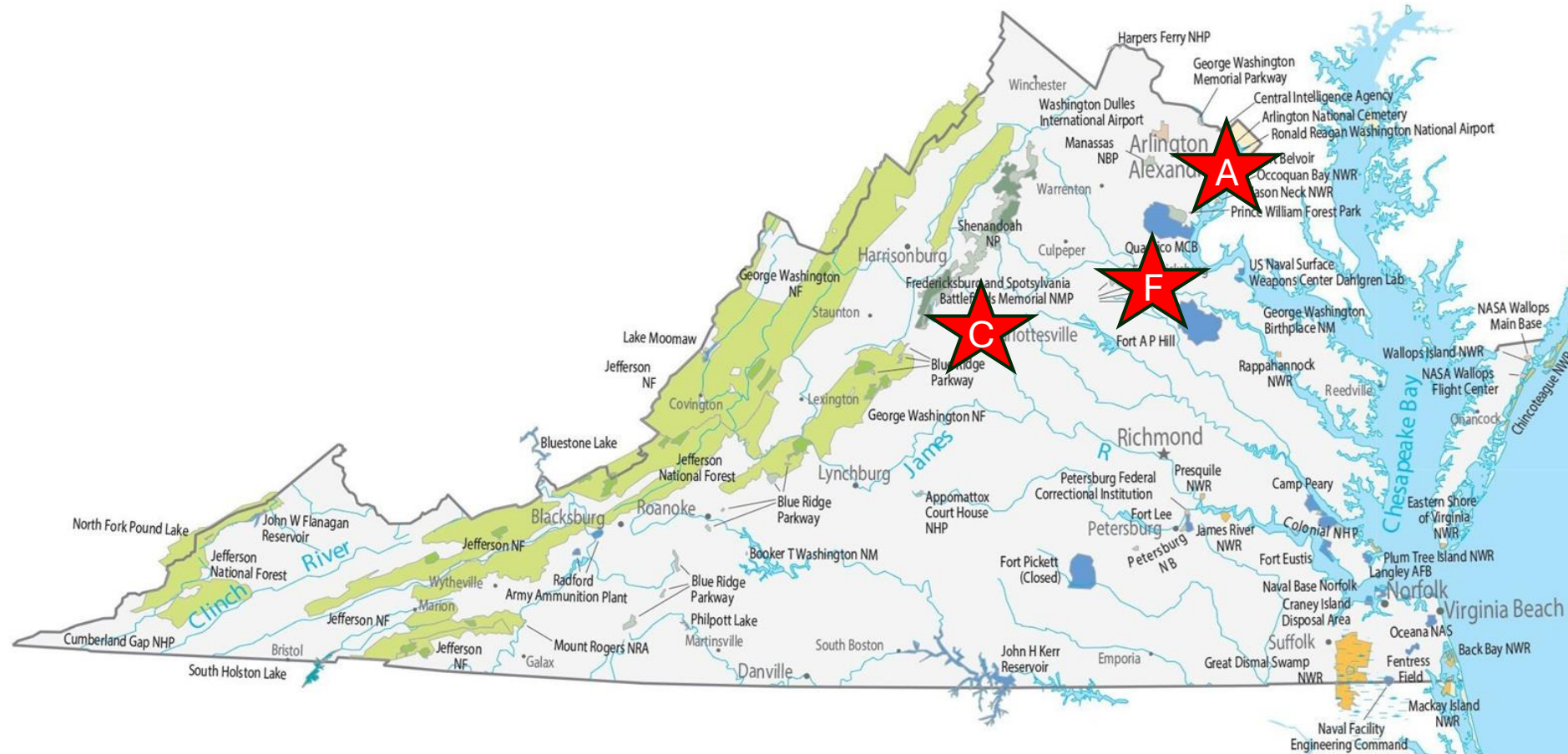
Scenario 2



Scenario 2 - Introduction

- In this scenario we have a multi-location rental car business
 - Three locations: Fredericksburg, Charlottesville, and Arlington
 - Three car classes: Compact, Midsize, and Luxury
 - Initial inventories of each class at each location
 - We can transfer cars from one location to another
 - We can capture more value of demand spikes at specific locations
 - Pre-position additional cars there – makes capacity more flexible than at hotels, for ex.
 - In this case, we will assume a fixed cost per transfer – done overnight
 - How do we position inventory and allocate our fleet to the booking slots to maximize revenue?
-

Multiple Locations – Fredericksburg, Charlottesville, and Arlington



Scenario 2 - Framing

Known Entities

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)

L = Set of rental car station locations

S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon

p_k = Price per instance of car rented for slot k

d_k = Demand for slot k

$c_{hii'}$ = Transfer cost per car of class h from location i to i'

D_{ij} = Set of booking slots with cars departing from location i on day j

A_{ij} = Set of booking slots with cars arriving at location i on day j

I_{hi0} = Initial inventory of car class h at location i

v_{hk} = 1 if car class h can satisfy demand for slot k , 0 otherwise

Decision Variables

x_{hk} = Number of cars in class h used to fill slot k

I_{hij} = Inventory of car class h at location i at end of business day j

$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j

Scenario 2 - Modeling

- Revised Objective (1): Maximize total revenue – total cost
 - Total Revenue = (Slot rental price × Cars assigned) summed over all car classes and slots
 - Total Cost = (Transfer cost × Cars transferred from location i to i') summed over all location pairs i and i' over all nights 0 to $\Delta-1$
 - Night 0 is the night immediately prior to the first day, and $\Delta-1$ is the one immediately prior to the last day

$$\max \sum_{h \in C} \sum_{k \in S} p_k x_{hk} - \sum_{h \in C} \sum_{i \in L} \sum_{i' \in L} \sum_{j=0}^{\Delta-1} c_{hii'} t_{hii'j}$$

Scenario 2 – Modeling (cont.)

- Inventory flow constraint (4) must be revised to accommodate transfers and multiple locations
- Modified Constraint (4): For a given car class h at location i on day j :
 - Ending Inventory = (Previous Day's Ending Inventory) – (Total Cars Transferred Out Previous Night) + (Total Cars Transferred In Previous Night) – (Cars Rented Out) + (Cars Returned)
 - $$I_{hij} = I_{hij-1} - \sum_{i' \in L} t_{hii'j-1} + \sum_{i' \in L} t_{hi'i j-1} - \sum_{k \in D_{ij}} x_{hk} + \sum_{k \in A_{ij}} x_{hk} \quad \forall h \in C; i \in L; j = 1.. \Delta$$

Scenario 2 – Modeling (cont.)

- New Constraint (5): For each car class h at each location i on each night j
 - Ensure total cars transferred out do not exceed the inventory
 - $\sum_{i' \in L} t_{hii'j} \leq I_{hij} \quad \forall h \in C; i \in L; j = 0.. \Delta - 1$
- Revise variable domain constraints (6 and 7) to include new variables
 - $I_{hij} \geq 0$ and integer $\forall h \in C; i \in L; j = 1.. \Delta$
 - $x_{hk}, t_{hii'j} \geq 0$ and integer $\forall h \in C; i, i' \in L; j = 0.. \Delta - 1$

Scenario 2 – Additional Remarks

- In this scenario we used a simplified transfer approach
 - In the next we will make it more realistic
- Let's take a minute to look at scenario 2 in the notebook

Breakout Session

- Implement model in notebook section 2.1
 - Complete TODO's
- Run and see results and optimal plan in 2.4-5

Multi-Location Rental Car Revenue Management with Realistic Transfers

Scenario 3



Scenario 3

Introduction

- In the previous scenario we used an economically unrealistic approach
 - Fixed price per car, unlimited transfers
- Truck-based transfers
 - More economically viable, realistic
 - Fixed number of trucks, capacity per truck
 - Cost per truck trip - can vary according to distance between locations
- Maintain an inventory of trucks
 - Each night, trucks can make one trip from their current location to another
 - Can transfer up to 8 cars
- Image source: <https://www.lohr.fr/general-automotive/>

Scenario 3 - Framing

Known Entities

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)

L = Set of rental car station locations

S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon

p_k = Price per instance of car rented for slot k

d_k = Demand for slot k

$c_{ii'}$ = Transfer cost per truck from location i to i'

D_{ij} = Set of booking slots with cars departing from location i on day j

A_{ij} = Set of booking slots with cars arriving at location i on day j

I_{hi0} = Initial inventory of car class h at location i

$v_{hk} = 1$ if car class h can satisfy demand for slot k , 0 otherwise

T_{i0} = Initial inventory of transport trucks at location i at end of day j

K^T = Capacity (cars) per transport truck

Decision Variables

x_{hk} = Number of cars in class h used to fill demand slot k

$y_{ii'j}$ = Number of transport trucks sent from location i to i' on night j

I_{hij} = Inventory of car class h at location i at end of day j

$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j

T_{ij} = Inventory of transport trucks at location i at end of day j

Scenario 3 - Modeling

- Revised Objective (1): Maximize total revenue – total cost
 - Total revenue same as before
 - Total cost does not need to sum over car classes anymore, since cost is per truck
 - $$\max \sum_{h \in C} \sum_{k \in S} p_k x_{hk} - \sum_{i \in L} \sum_{i' \in L} \sum_{j=0}^{\Delta-1} c_{ii'} y_{ii'j}$$
- Revised Constraint (5): For each location i on each night j
 - Ensure total trucks transferred out do not exceed the inventory
 - $$\sum_{i' \in L} y_{ii'j} \leq T_{ij} \quad \forall i \in L; j = 0.. \Delta - 1$$

Scenario 3 – Modeling: Your Turn! (Time Permitting)

- New Constraint (6): For each location i on days $j = 1..\Delta$ over the horizon:
 - Develop a constraint to ensure correct ending truck inventory for each day
 - Remember, a single trip per night – trucks stay put after delivering
 - Use the car inventory flow balance constraint as a similar example
- New Constraint (7): For each pair of locations i and i' over nights $j = 0..\Delta - 1$:
 - Ensure total cars transported from i to i' does not exceed capacity of trucks going on this route
- Breakout Session: Formulate these on paper first (10 minutes)

Scenario 3 – Modeling (cont.)

- New Constraint (6): For each location i on days $j = 1..\Delta$ over the horizon:
 - Ending Truck Inventory = Previous Day's Ending Truck Inventory – Trucks Transferred Out + Trucks Transferred In
 - $T_{ij} = T_{ij-1} - \sum_{i' \in L} y_{ii'j-1} + \sum_{i' \in L} y_{i'i j-1} \quad \forall i \in L; j = 1..\Delta$
- New Constraint (7): For each pair of locations i and i' over nights $j = 0..\Delta - 1$:
 - Ensure total cars transported from i to i' does not exceed capacity of trucks going on this route
 - $\sum_{h \in C} t_{hii'j} \leq K^T y_{ii'j} \quad \forall i, i' \in L; j = 0..\Delta - 1$

Scenario 3 – Modeling (cont.)

- Domains for T variables:
 - $I_{hij}, T_{ij} \geq 0$ and integer $\forall h \in C; i \in L; j = 1.. \Delta$

Scenario 3 – Additional Remarks

- In this scenario we incorporated more realistic truck transfers
 - Simple policy – assumes a single one-way trip per night
 - It's also possible that trucks visit multiple locations in a single night
 - Need to incorporate the transfer policy into the constraints
- Let's take a minute to explore scenario 3 in the notebook

Breakout Session

- Implement model in notebook section 3.1
 - Complete TODO's
- Run and see results and optimal plan in 3.4-5

Incorporating Dynamic Pricing and Fluctuating Demand

Scenario 4



Scenario 4 - Introduction

- In previous scenarios we made some highly unrealistic assumptions
 - Forecasted demand and price were accurate and not subject to change
 - Unforeseen events can quickly change demand and price
 - Taylor Swift makes a surprise visit in 2 days
 - Also, forecasts tend to be more accurate in the near-term
 - Should periodically update and adjust capacity allocation accordingly
 - Need to honor reservations already sold (per overbooking policy)
 - How to handle this?
 - Integrate committed demand into decision making
 - Employ a rolling horizon and re-solve/re-optimize continuously
-

Scenario 4 - Framing

Known Entities

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)

L = Set of rental car station locations

S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon

p_k = Price per instance of car rented for demand slot k

d_k = Demand for slot k

d_k^c = Committed demand for slot k

$c_{ii'}$ = Transfer cost per truck from location i to i'

D_{ij} = Set of booking slots with cars departing from location i on day j

A_{ij} = Set of booking slots with cars arriving at location i on day j

I_{hi0} = Initial inventory of car class h at location i

$u_{kk'}$ = 1 if booking slots k and k' conflict, 0 otherwise

v_{hk} = 1 if car class h can satisfy demand for slot k , 0 otherwise

T_{i0} = Initial inventory of transport trucks at location i at end of day j

K^T = Capacity (cars) per transport truck

Decision Variables

x_{hk} = Number of cars in class h used to fill demand slot k

$y_{ii'j}$ = Number of transport trucks sent from location i to i' on night j

I_{hij} = Inventory of car class h at location i at end of day j

$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j

T_{ij} = Inventory of transport trucks at location i at end of day j

Scenario 4 - Modeling

- One simple constraint needs to be added
 - Ensure we meet committed demand for all slots
 - Modify constraint 2 and add a "part A" to ensure enough cars are allocated to meet committed demand:
 - $d_k^c \leq \sum_{h \in C} x_{hk} \leq d_k \quad \forall k \in S$
 - We can implement this in Python as two separate constraints
- Take a minute to look at scenario 4 in the notebook

Wrapping Up: Ethical Use of Revenue Management

- RM can be used in genuinely unethical ways
 - Exploit through unjustly high costs for essential services
 - Shut some groups out of access to goods or services necessary for survival
 - Medical care, housing, etc.
 - Industry collusion leading to monopolistic exploitation (e.g. RealPage lawsuit)
- RM can also be used in ways that create perceived unfairness
 - Why did that person only pay \$X when I had to pay \$Y?
 - Happens when customers can't perceive reason for price differences
 - Pricing appears arbitrary and exploitative
 - Leads to customer resentment and eventual churn
- Some questions to guide ethical and responsible use of RM
 - Am I dealing with goods or services necessary for survival and the possibility of shutting people out altogether?
 - Is there a clearly-perceived reason for price distinctions? Do they enable me to serve different segments?
 - Ethical Example: Airlines
 - If all seats sold 6 months in advance at low price, no availability for business flyers
- Bottom Line: Don't try to gain the world at the cost of your soul



Thanks for attending!

**Questions and Further
Discussion**